THE UNITED REPUBLIC OF TANZANIA

ORGANISATION OF HEADS OF NON-GOVERNMENT SECONDARY SCHOOLS

(OHONGSS)

KIGOMA REGIONS

FORM FOUR MOCK EXAMINATION

(ACTUAL PRACTICAL)

032/2A **CHEMISTRY 2A**

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Time: 2:30 HOURS **YEAR: 2025**

INSTRUCTIONS

1. This paper consists of two (2) questions. Answer all questions.

2. Each question carries twenty five (25) marks.

3. All writings should be in blue or black ink, except for diagrams which must be in pencil.

4. Cellular phones and any unauthorized are not allowed in the examination room.

5. Write your Examination number on every page of your answer booklets

6. You may use the following atomic masses

H=1, O=16, Na=23 and Cl=35.5

1. You are provided with the following

BB: 0.25M sodium thiosulphate ($Na_2S_2O_3$)

DD: 2M hydrochloric acid (HCl)

Stop watch

Distilled water

White paper with a mark X

Procedures

(i)Using measuring cylinder, measure $5cm^3$ of solution DD and put into $100cm^3$ beaker and palaces the beaker on the mark X

(ii) Measure $50cm^3$ of solution BB and put into $100cm^3$ containing solution DD and immediately start the stop watch

(iii) Switch off the stop watch when the mark X disappears and record the time taken for mark X to disappear.

(iv)Repeat procedures (i) to (ii) using the data shown in the table

Experiment number	Volume of DD(cm^3)	Volume of BB(cm^3)	Volume of distilled water(cm^3)	Time, t(s)	Rate 1/t (s^{-1})
1	5	50	0		
2	5	40	10		
3	5	30	20		
4	5	20	30		
5	5	10	40		

Questions

(a) Complete the table above

(b)Write a balanced question for the reaction between **BB** and **DD**

(C) What substance was produced during the reaction which obscured the letter X?

(d) Plot the graph of the volume of sodium thiosulphate against the rate.

(e) Use the graph to explain how variation of concentration affects the rate of chemical reaction.

2. Sample **Q** is a simple salt containing one cation and one anion. Carry out the experiments described below. Record carefully your observations, inferences and finally identify the anion and cation present in sample **Q**

S/n	Experiment	Observation	Inference
(a)	Observe the appearance of sample Q		
(b)	Dissolve a little amount of sample Q in distilled water in a test tube stir and then boil		
(c)	Put a spatulaful of sample Q in a test tube .add concentrated sulphuric acid and warm.		
(d)	Put a spatulaful of sample Q in a test tube .add dilute nitric acid .divide the resulting solution into three portions (I) To the first portion add NaOH solution till in excess		
	(ii) To the second portion add KI solution till in excess. Warm and cool the mixture		
	(iii) To the third portion add $AgNO_3$ solution followed by dilute HNO_3 and the NH_3 solution		

Conclusion

(a) The cation in sample Q is.....

(b)The anion in sample Q is.....

(C)The formula of a compound Q is.....

(d) The name of the compound Q is.....